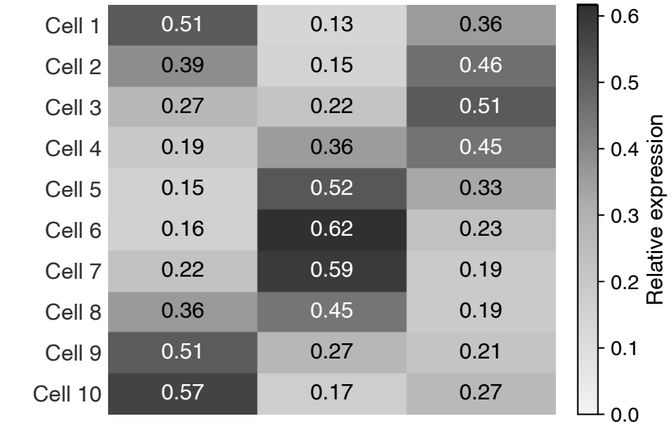
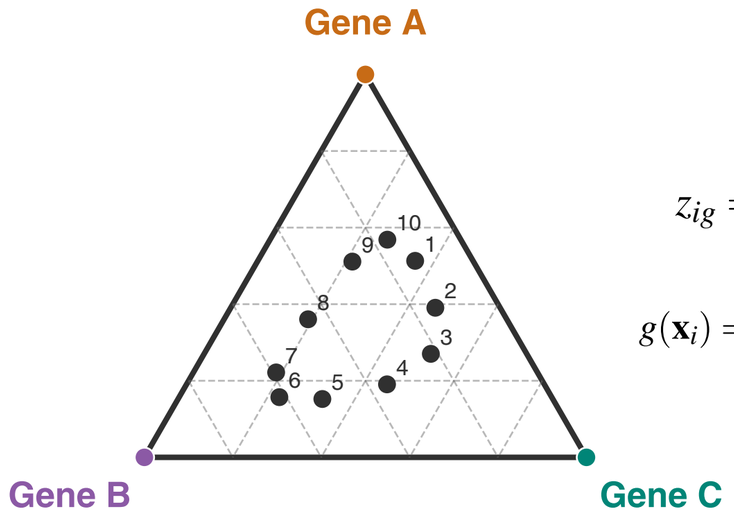
1. Gene expression in CLR space

	Gene A	Gene B	Gene C
Cell 1	0.51	0.13	0.36
Cell 2	0.39	0.15	0.46
Cell 3	0.27	0.22	0.51
Cell 4	0.19	0.36	0.45
Cell 5	0.15	0.52	0.33
Cell 6	0.16	0.62	0.23
Cell 7	0.22	0.59	0.19
Cell 8	0.36	0.45	0.19
Cell 9	0.51	0.27	0.21
Cell 10	0.57	0.17	0.27



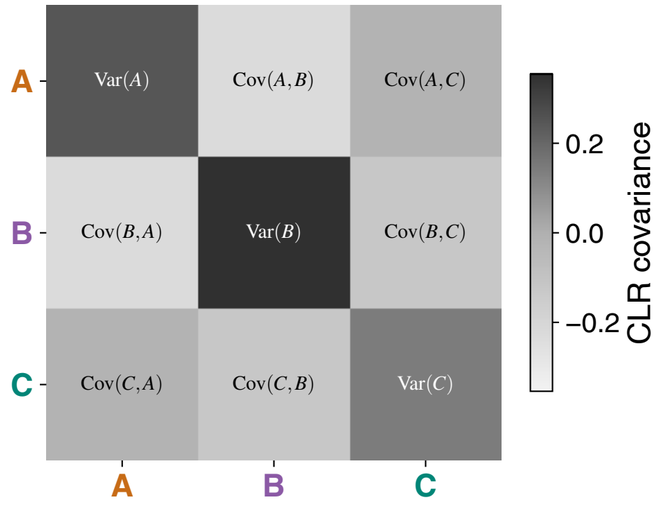


$$z_{ig} = \log \frac{x_{ig}}{g(\mathbf{x}_i)}$$

$$g(\mathbf{x}_i) = \left(\prod_g x_{ig}\right)^{1/G}$$

2. Gene covariance

Σ_{genes} in CLR space



3. Maximize first Fourier variance

$$F(\pi) = \mathbf{c}^T \Sigma_\pi \mathbf{c} + \mathbf{s}^T \Sigma_\pi \mathbf{s}$$

π: candidate gene order

Σ_π: CLR covariance reordered by π

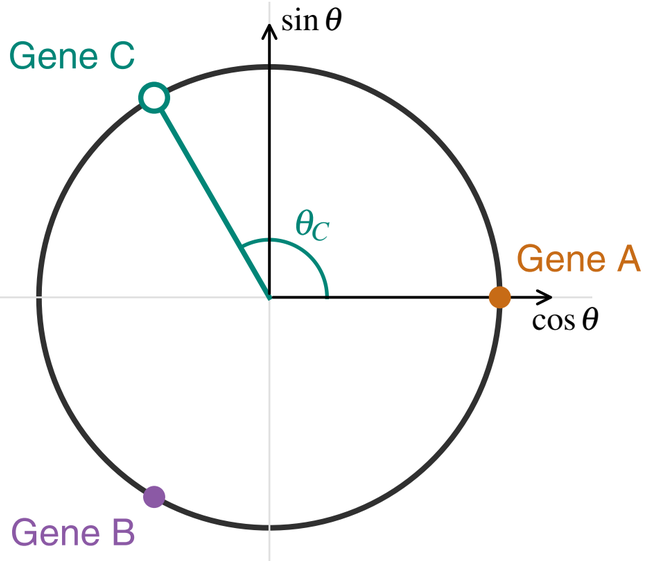
$$\theta_g = 2\pi(g-1)/G$$

$$c_g = \sqrt{\frac{2}{G}} \cos\theta_g, \qquad s_g = \sqrt{\frac{2}{G}} \sin\theta_g$$

$$\boxed{\hat{\pi} = \arg\max_\pi \; F(\pi)}$$

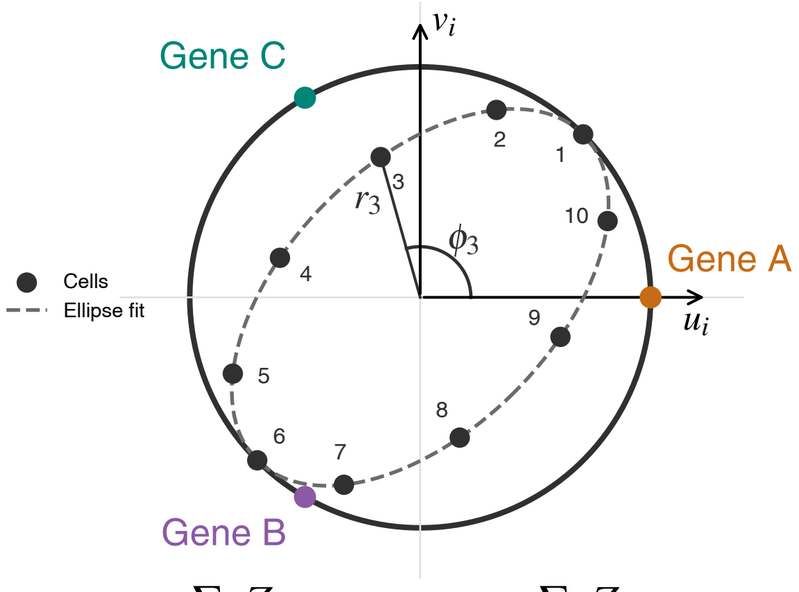
$\hat{\pi}$ is the optimal gene order

4. Gene ordered circular coordinates



$$\theta_C = \frac{2\pi(2-1)}{3} = \frac{2\pi}{3}$$

5. Project cells onto joint feature-sample geometry



$$u_i = \sum_g Z_{i,\hat{\pi}(g)} c_g \qquad v_i = \sum_g Z_{i,\hat{\pi}(g)} s_g$$